

HW 2 Cross-Validation in Sports Data

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Sport & Research Topic:

Soccer: How Does the Possession of the Players & Squads Affect the Offensive Performance in the Premier League

Introduction:

This study examines how advanced possession metrics predict offensive performance in the English Premier League. Using FBref data from 2017–2018 to 2024–2025, I analyze both player- and team-level indicators. Offensive performance is measured as the combined **expected goals and assists per 90 minutes (xG+xAG)**, while **predictors cover four areas: Touches, Take-Ons, Carriers, and Receiving**. Recent work by Link and Hoernig (2017) and Casal et al. (2019) shows that detailed ball possession analysis—integrating both quantitative and qualitative aspects—yields critical tactical insights. I hypothesize that refined possession metrics are positively correlated with improved offensive outcomes at both levels.

Methods:

Preprocessing: The data were cleaned (infinities replaced, missing values imputed), column names were standardized, and data was scaled using StandardScaler

Player-Level Analysis (Linear Regression + Validation Set):

- **Target Variable:** xG+xAG per 90 minutes.
- **Predictors:** Player-level advanced possession metrics & Positions.
- **Validation:** A 75/25 validation set; MSE of 0.04415 and R^2 of 0.36435, suggesting that while the predictions are numerically close to the observed values, approximately 36% of the variance in xG+xAG is explained by the predictors.

Player-Level Analysis (Logistic Regression + K-Fold):

- **Target Variable:** Binary indicator (1 = high impact if xG+xAG > median; 0 = low impact)
- **Predictors:** Player-level advanced possession metrics & Positions.
- **Validation:** Using a stratified 5-fold model has an accuracy of 86.6%, with high precision (88.0%) and recall (83.3%) resulting in an F1 score of 85.6%. Additionally, a ROC-AUC of 86.5% indicates a strong overall ability to discriminate between the classes across different thresholds, indicating that Possession metrics robustly classify players' offensive impact.

Squad-Level Analysis (Multi-Class Logistic Regression + LOOCV):

- **Target Variable:** Multi-class performance (0 = low, 1 = medium, 2 = high; based on xG+xAG quantiles)
- **Predictors:** Team-level Advanced possession metrics & Possession
- **Validation:** Accuracy of 75%; Class-specific precision: 61%–86%; The confusion matrix for LOOCV was: [[41, 12, 0], [11, 35, 7], [0, 10, 44]], indicating while the model reliably differentiates between team performance tiers, its precision varies across the different classes.